



K1 CSE3023 PEMBANGUNAN APLIKASI BERASASKAN WEB

SEM 4

LAB REPORT 7

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NO. MATRIC:

S75776

LAB EXERCISE 7

index.html

Welcome to the Student Portal

Click below to sign in and access the Subject Management System module.

[Go to Login Portal](#)

login.jsp

Student Login

Matric No:

Password:

[Login](#)

Don't have an account yet? [Register here](#)

register.html

Create Student Account

Matric No:

Password:

Full Name:

Register

Cancel

dashboard.jsp

Welcome back, Student!

Logout

Logged in Matric No: **S75776**

Academic Management Module

Click below to register new semester courses, modify existing details, or view current enrollment schedules.

Manage Registered Subjects

SubjectServlet

Registered Subjects List

Logged in User (Matric No): S75776

[+ Register New Subject](#)

System ID	Subject Code	Subject Name	Actions
2	cse3023	Web Application Development	Edit Delete

[← Return to Dashboard](#)

registerSubject.jsp

Register New Subject

Subject Code:

e.g., CSE3023

Subject Name:

e.g., Web Application Development

Register

Cancel

Modify Subject

Subject Code:

Subject Name:

Save Changes

Cancel

Source code

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Portal Gateway</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; margin-top: 100px; background-color: #f4f7f6; }
    .welcome-box { background: white; padding: 40px; display: inline-block; border-radius: 8px; box-shadow: 0 4px 10px rgba(0,0,0,0.05); border-top: 5px solid #2ecc71; }
    h1 { color: #2c3e50; }
    .enter-btn { display: inline-block; background-color: #2ecc71;
               color: white; padding: 12px 25px; text-decoration: none; border-radius: 4px;
               font-weight: bold; font-size: 16px; margin-top: 20px; }
  </style>
</head>
<body>
  <div class="welcome-box">
    <h1>Welcome to the Student Portal</h1>
    <p>Click below to sign in and access the Subject Management System module.</p>
    <a href="login.jsp" class="enter-btn">Go to Login Portal</a>
  </div>
</body>
</html>
```

Login.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <title>Student Portal Login</title>
  <style>
    body { font-family: Arial, sans-serif; background-color: #f4f7f6; display: flex; justify-content: center; align-items: center; height: 100vh; margin: 0; }
    .login-container { background: white; padding: 30px; border-radius: 8px; box-shadow: 0 4px 15px rgba(0,0,0,0.1); width: 350px; border-top: 5px solid #2ecc71; }
    h2 { text-align: center; color: #2c3e50; margin-bottom: 20px; }
    label { font-weight: bold; color: #34495e; }
    input[type="text"], input[type="password"] { width: 100%; padding: 10px; margin: 8px 0 20px 0; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
    input[type="submit"] { width: 100%; background-color: #2ecc71; color: white; padding: 12px; border: none; border-radius: 4px; cursor: pointer; font-weight: bold; font-size: 16px; }
    .msg { color: #e74c3c; text-align: center; font-weight: bold; margin-bottom: 15px; }
    .register-prompt { text-align: center; margin-top: 20px; font-size: 14px; color: #7f8c8d; border-top: 1px solid #eee; padding-top: 15px; }
    .register-prompt a { color: #2ecc71; text-decoration: none; font-weight: bold; }
    .register-prompt a:hover { text-decoration: underline; }
  </style>
</head>
<body>
```

```

<body>
  <div class="login-container">
    <h2>Student Login</h2>

    <!-- Displays status alerts (e.g., success message after registering successfully) -->
    <%
      String msg = request.getParameter("msg");
      if(msg != null) {
    %>
    <div class="msg" style="color: #27ae60;"><%= msg %></div>
    <% } %>

    <form action="doLogin.jsp" method="post">
      <label>Matric No:</label>
      <input type="text" name="username" placeholder="Enter your Matric No" required>

      <label>Password:</label>
      <input type="password" name="password" placeholder="Enter your Password" required>

      <input type="submit" value="Login">
    </form>

    <div class="register-prompt">
      Don't have an account yet? <a href="register.html">Register here</a>
    </div>
  </div>
</body>
</html>

```

doLogin.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.sql.*"%>
<%
  String user = request.getParameter("username");
  String pass = request.getParameter("password");

  Connection conn = null;
  PreparedStatement ps = null;
  ResultSet rs = null;

  try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    conn = DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db", "root", "");

    // Accurate table lookup pattern
    String sql = "SELECT * FROM students WHERE matric_no=? AND password=?";
    ps = conn.prepareStatement(sql);
    ps.setString(1, user);
    ps.setString(2, pass);

    rs = ps.executeQuery();

    if (rs.next()) {
      session.setAttribute("sess_user", rs.getString("matric_no")); // Set context variable
      session.setAttribute("sess_fname", rs.getString("fullname"));

      response.sendRedirect("dashboard.jsp");
    } else {
      response.sendRedirect("login.jsp?msg=Invalid username or password..!");
    }
  } catch (Exception e) {
    out.println("Database Error: " + e.getMessage());
  } finally {
    if (rs != null) try { rs.close(); } catch (Exception e) {}
    if (ps != null) try { ps.close(); } catch (Exception e) {}
    if (conn != null) try { conn.close(); } catch (Exception e) {}
  }
}
%>

```

Register.html

```

<!DOCTYPE html>
<html>
<head>
  <title>Student Registration</title>
  <style>
    body { font-family: Arial, sans-serif; background-color: #f4f7f6; margin: 40px; text-align: center; }
    .form-card { background: white; padding: 25px; display: inline-block; border-radius: 8px; box-shadow: 0 4px 8px rgba(0,0,0,0.05); text-align: left; width: 350px; border-top: 3px solid #2c3e50; }
    h2 { color: #2c3e50; margin-top: 0; }
    label { font-weight: bold; display: block; margin-top: 10px; color: #34495e; }
    input[type="text"], input[type="password"] { width: 100%; padding: 8px; margin: 5px 0 15px 0; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
    .btn-submit { background-color: #2ecc71; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; font-weight: bold; width: 48%; }
    .btn-cancel { background-color: #95a5a6; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; font-weight: bold; width: 48%; float: right; }
  </style>
</head>
<body>
  <div class="form-card">
    <h2>Create Student Account</h2>
    <form action="UserServlet" method="post">
      <label>Matric No:</label>
      <input type="text" name="matricNo" required>

      <label>Password:</label>
      <input type="password" name="password" required>

      <label>Full Name:</label>
      <input type="text" name="fullname" required>

      <input type="submit" value="Register" class="btn-submit">
      <a href="login.jsp" class="btn-cancel">Cancel</a>
    </form>
  </div>
</body>
</html>

```

dashboard.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
// Protection Layer: Kick out users who try to deep-link to the dashboard without logging in
String user = (String) session.getAttribute("sess_user");
if(user == null) {
    response.sendRedirect("login.jsp?msg=Please login first.");
    return;
}
%>

<!DOCTYPE html>
<html>
<head>
<title>Student Dashboard</title>
<style>
body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f7f6; }
.dashboard-card { background: white; padding: 30px; border-radius: 8px; box-shadow: 0 4px 8px rgba(0,0,0,0.05); max-width: 600px; margin: 0 auto; }
h1 { color: #2c3e50; }
.btn-link { display: inline-block; background-color: #3498db; color: white; padding: 12px 24px; text-decoration: none; border-radius: 4px; font-weight: bold; margin-top: 10px; }
.btn-logout { display: inline-block; background-color: #e74c3c; color: white; padding: 8px 15px; text-decoration: none; border-radius: 4px; float: right; font-size: 14px; }
</style>
</head>
<body>
<div class="dashboard-card">
<a href="logout.jsp" class="btn-logout">Logout</a>
<h2>Welcome back, Student!</h2>
<p>Logged in Matric No: <strong><%= user %></strong></p>
<hr style="border:0; border-top: 1px solid #eee; margin: 20px 0;">
<h3>Academic Management Module</h3>
<p>Click below to register new semester courses, modify existing details, or view current enrollment schedules.</p>
<a href="SubjectServlet?action=list" class="btn-link">Manage Registered Subjects</a>
</div>
</body>
</html>
```

registerSubject.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<title>Register New Subject</title>
<style>
body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f7f6; }
.form-container { width: 400px; background: white; padding: 20px; border-radius: 8px; box-shadow: 0 4px 8px rgba(0,0,0,0.05); border-top: 5px solid #2ecc71; }
h2 { color: #2c3e50; margin-top: 0; }
input[type="text"] { width: 100%; padding: 8px; margin: 10px 0 20px 0; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
input[type="submit"] { background-color: #2ecc71; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }
input[type="button"] { background-color: #95a5a6; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; margin-left: 10px; }
</style>
</head>
<body>
<div class="form-container">
<h2>Register New Subject</h2>
<form action=".."SubjectServlet?action=insert" method="post">
<label>Subject Code:</label>
<input type="text" name="subject_code" placeholder="e.g., CSE3023" required>
<label>Subject Name:</label>
<input type="text" name="subject_name" placeholder="e.g., Web Application Development" required>
<input type="submit" value="Register">
<input type="button" value="Cancel" onclick="window.location.href='.."SubjectServlet?action=list"'>
</form>
</div>
</body>
</html>
```

updateSubject.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="com.lab.bean.SubjectBean"%>
<!DOCTYPE html>
<html>
<head>
<title>Modify Subject Details</title>
<style>
body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f7f6; }
.form-container { width: 400px; background: white; padding: 20px; border-radius: 8px; box-shadow: 0 4px 8px rgba(0,0,0,0.05); border-top: 5px solid #3498db; }
h2 { color: #2c3e50; margin-top: 0; }
input[type="text"] { width: 100%; padding: 8px; margin: 10px 0 20px 0; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
input[type="submit"] { background-color: #3498db; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }
input[type="button"] { background-color: #95a5a6; color: white; padding: 10px 15px; border: none; border-radius: 4px; cursor: pointer; margin-left: 10px; }
</style>
</head>
<body>
<%
SubjectBean subject = (SubjectBean) request.getAttribute("subject");
if(subject != null) {
%>
<div class="form-container">
<h2>Modify Subject</h2>
<form action="SubjectServlet?action=update" method="post">
<input type="hidden" name="id" value="<%= subject.getId() %>">
<label>Subject Code:</label>
<input type="text" name="subject_code" value="<%= subject.getSubject_code() %>" required>
<label>Subject Name:</label>
<input type="text" name="subject_name" value="<%= subject.getSubject_name() %>" required>
<input type="submit" value="Save Changes">
<input type="button" value="Cancel" onclick="window.location.href='SubjectServlet?action=list'">
</form>
</div>
```

```

</div>
<%> } else { %>
<p style="color:red; font-weight:bold;">Error: Unable to load the requested subject data record.</p>
<%> } %>
</body>
</html>

```

viewSubject.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.List"%>
<%@page import="com.lab.bean.SubjectBean"%>
<!DOCTYPE html>
<html>
<head>
<title>My Registered Subjects</title>
<style>
body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f7f6; }
table { width: 100%; border-collapse: collapse; margin-top: 20px; background: white; }
th, td { padding: 12px; text-align: left; border-bottom: 1px solid #ddd; }
th { background-color: #2ecc71; color: white; }
tr:hover { background-color: #f1f1f1; }
.btn-add { background-color: #3498db; color: white; padding: 10px 15px; text-decoration: none; border-radius: 4px; display: inline-block; margin-bottom: 15px; font-weight: bold; }
.btn-edit { color: #3498db; text-decoration: none; margin-right: 15px; font-weight: bold; }
.btn-delete { color: #e74c3c; text-decoration: none; font-weight: bold; }
</style>
</head>
<body>

<body>

<h2>Registered Subjects List</h2>
<p>Logged in User (Matric No): <strong><%= session.getAttribute("sess_user") %></strong></p>

<a href="subject/registerSubject.jsp" class="btn-add">+ Register New Subject</a>

<table>
<tr>
<th>System ID</th>
<th>Subject Code</th>
<th>Subject Name</th>
<th>Actions</th>
</tr>
<%
List<SubjectBean> listSubject = (List<SubjectBean>) request.getAttribute("listSubject");
if (listSubject != null && !listSubject.isEmpty()) {
for (SubjectBean subject : listSubject) {
%>
<tr>
<td><%= subject.getId() %></td>
<td><%= subject.getSubject_code() %></td>
<td><%= subject.getSubject_name() %></td>
<td>
<a href="SubjectServlet?action=editid=<%= subject.getId() %>" class="btn-edit">Edit</a>
<a href="SubjectServlet?action=deleteid=<%= subject.getId() %>" class="btn-delete" onclick="return confirm('Are you sure you want to drop this subject?')">Delete</a>
</td>
</tr>
<%>
}
} else {
<tr>
<td colspan="4" style="text-align: center; color: #7f8c8d; padding: 20px;">No registered subjects found for your account.</td>
</tr>
<%>
}
</body>

<a href="dashboard.jsp" style="text-decoration: none; color: #7f8c8d;">&larr; Return to Dashboard</a>

```

StudenBean.java


```

package com.lab.bean;
import java.io.Serializable;

public class StudentBean implements Serializable {
    private int id;
    private String matricNo; // Exact match to column 'matricNo'
    private String password;
    private String fullname;

    public StudentBean() {}

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getMatricNo() { return matricNo; }
    public void setMatricNo(String matricNo) { this.matricNo = matricNo; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }

    public String getFullname() { return fullname; }
    public void setFullname(String fullname) { this.fullname = fullname; }

}

```

SubjectBean.java

```

package com.lab.bean;
import java.io.Serializable;

public class SubjectBean implements Serializable {
    private int id;
    private String matric_no; // Exact match to registered_subjects 'matric_no'
    private String subject_code;
    private String subject_name;

    public SubjectBean() {}

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getMatric_no() { return matric_no; }
    public void setMatric_no(String matric_no) { this.matric_no = matric_no; }

    public String getSubject_code() { return subject_code; }
    public void setSubject_code(String subject_code) { this.subject_code = subject_code; }

    public String getSubject_name() { return subject_name; }
    public void setSubject_name(String subject_name) { this.subject_name = subject_name; }

}

```

SubjectServlet.java

```
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    HttpSession session = request.getSession();
    String currentMatric = (String) session.getAttribute("sess_user");

    if (currentMatric == null) {
        response.sendRedirect("login.jsp?msg=Authentication required.");
        return;
    }

    String action = request.getParameter("action");
    if (action == null) { action = "list"; }

    try {
        switch (action) {
            case "insert":
                insertSubject(request, response, currentMatric);
                break;
            case "delete":
                deleteSubject(request, response);
                break;
            case "edit":
                showEditForm(request, response);
                break;
            case "update":
                updateSubject(request, response);
                break;
            default:
                listSubjects(request, response, currentMatric);
                break;
        }
    }

    } catch (Exception ex) {
        throw new ServletException(ex);
    }
}

private void listSubjects(HttpServletRequest request, HttpServletResponse response, String currentMatric)
    throws Exception {
    List<SubjectBean> listSubject = subjectDAO.getSubjectsByMatric(currentMatric);
    request.setAttribute("listSubject", listSubject);
    request.getRequestDispatcher("subject/viewSubjects.jsp").forward(request, response);
}

private void insertSubject(HttpServletRequest request, HttpServletResponse response, String currentMatric)
    throws Exception {
    String code = request.getParameter("subject_code");
    String name = request.getParameter("subject_name");

    SubjectBean newSubject = new SubjectBean();
    newSubject.setMatric_no(currentMatric); // Saves to registered_subjects.matric_no
    newSubject.setSubject_code(code);
    newSubject.setSubject_name(name);

    subjectDAO.addSubject(newSubject);
    response.sendRedirect("SubjectServlet?action=list");
}

private void showEditForm(HttpServletRequest request, HttpServletResponse response)
    throws Exception {
    int id = Integer.parseInt(request.getParameter("id"));
    SubjectBean existingSubject = subjectDAO.getSubjectById(id);
    request.setAttribute("subject", existingSubject);
    request.getRequestDispatcher("subject/updateSubject.jsp").forward(request, response);
}
```

```

private void updateSubject(HttpServletRequest request, HttpServletResponse response)
    throws Exception {
    int id = Integer.parseInt(request.getParameter("id"));
    String code = request.getParameter("subject_code");
    String name = request.getParameter("subject_name");

    SubjectBean subject = new SubjectBean();
    subject.setId(id);
    subject.setSubject_code(code);
    subject.setSubject_name(name);

    subjectDAO.updateSubject(subject);
    response.sendRedirect("SubjectServlet?action=list");
}

private void deleteSubject(HttpServletRequest request, HttpServletResponse response)
    throws Exception {
    int id = Integer.parseInt(request.getParameter("id"));
    subjectDAO.deleteSubject(id);
    response.sendRedirect("SubjectServlet?action=list");
}

@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    HttpSession session = request.getSession();
    String currentMatric = (String) session.getAttribute("sess_user");

    if (currentMatric == null) {
        response.sendRedirect("login.jsp?msg=Authentication required.");
        return;
    }

    String action = request.getParameter("action");

    try {

        if ("insert".equals(action)) {
            insertSubject(request, response, currentMatric);
        } else if ("update".equals(action)) {
            updateSubject(request, response);
        } else {
            response.sendRedirect("SubjectServlet?action=list");
        }

    } catch (Exception ex) {
        throw new ServletException(ex);
    }
}

```

UserServlet.java

```

@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    String matricNo = request.getParameter("matricNo");
    String password = request.getParameter("password");
    String fullname = request.getParameter("fullname");

    StudentBean newStudent = new StudentBean();
    newStudent.setMatricNo(matricNo);
    newStudent.setPassword(password);
    newStudent.setFullname(fullname);

    boolean success = studentDAO.registerStudent(newStudent);

    if (success) {
        response.sendRedirect("login.jsp?msg=Registration successful! Please login.");
    } else {
        response.sendRedirect("register.html?msg=Registration failed. Matric No might exist.");
    }
}

```

StudentDAO.java

```

package com.lab.dao;
import java.sql.*;
import com.lab.bean.StudentBean;

public class StudentDAO {
    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db", "root", "");
    }

    public boolean registerStudent(StudentBean student) {
        String sql = "INSERT INTO students (matric_no, password, fullname) VALUES (?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
            ps.setString(1, student.getMatricNo());
            ps.setString(2, student.getPassword());
            ps.setString(3, student.getFullname());
            return ps.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }
}

```

SubjectDAO.java

```

package com.lab.dao;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
import com.lab.bean.SubjectBean;

public class SubjectDAO {
    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3307/lab7_db", "root", "");
    }

    public boolean addSubject(SubjectBean subject) {
        String sql = "INSERT INTO registered_subjects (matric_no, subject_code, subject_name) VALUES (?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
            ps.setString(1, subject.getMatric_no());
            ps.setString(2, subject.getSubject_code());
            ps.setString(3, subject.getSubject_name());
            return ps.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }
}

```

```

public List<SubjectBean> getSubjectsByMatric(String matric_no) {
    List<SubjectBean> list = new ArrayList<>();
    String sql = "SELECT * FROM registered_subjects WHERE matric_no = ?";
    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setString(1, matric_no);
        try (ResultSet rs = ps.executeQuery()) {
            while (rs.next()) {
                SubjectBean sb = new SubjectBean();
                sb.setId(rs.getInt("id"));
                sb.setMatric_no(rs.getString("matric_no"));
                sb.setSubject_code(rs.getString("subject_code"));
                sb.setSubject_name(rs.getString("subject_name"));
                list.add(sb);
            }
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

```

```

public SubjectBean getSubjectById(int id) {
    String sql = "SELECT * FROM registered_subjects WHERE id = ?";
    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setInt(1, id);
        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                SubjectBean sb = new SubjectBean();
                sb.setId(rs.getInt("id"));
                sb.setMatric_no(rs.getString("matric_no"));
                sb.setSubject_code(rs.getString("subject_code"));
                sb.setSubject_name(rs.getString("subject_name"));
                return sb;
            }
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return null;
}

public boolean updateSubject(SubjectBean subject) {
    String sql = "UPDATE registered_subjects SET subject_code = ?, subject_name = ? WHERE id = ?";
    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setString(1, subject.getSubject_code());
        ps.setString(2, subject.getSubject_name());
        ps.setInt(3, subject.getId());
        return ps.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

public boolean deleteSubject(int id) {
    String sql = "DELETE FROM registered_subjects WHERE id = ?";
    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(sql)) {
        ps.setInt(1, id);
        return ps.executeUpdate() > 0;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}
}

```